EXPERIMENT 2

AIM: Interfacing of LED with an embedded core (LPC 2148)

THEORY: The LPC2148 is a microcontroller based on the ARM7TDMI-S processor, created by NXP.It is commonly used in embedded systems because it includes various features like GPIO pins, timers, serial communication ports, ADC, and more.

An LED, or Light Emitting Diode, is a type of semiconductor that produces light when electricity flows through it in the correct direction.
This LED can be connected to the GPIO pins of the LPC2148 to show how digital output works.

The GPIO pins on the LPC2148 can be set up as either input or output.
For using the LED, the selected GPIO pin is set as an output.By sending a HIGH (1) or LOW (0) signal to that pin, the LED can be turned on or off, depending on how it is connected in the circuit.

In a standard LED setup, the LED is connected to a GPIO pin via a resistor that limits the current.
When the GPIO pin gives the right voltage, the LED gets powered and lights up.By changing the voltage level on the pin with a delay, the LED can be made to blink.

ALGORITHM:

1. Start the program.
2. Set up the LPC2148 microcontroller.
3. Choose the GPIO pin that is linked to the LED.
4. Set that GPIO pin to act as an output.
5. Send a high signal to the GPIO pin to switch the LED on.
6. Add a small delay.
7. Send a low signal to the GPIO pin to switch the LED off.
8. Add another small delay.

9. Keep repeating steps 5 to 8 to make the LED blink.
10. End the program.

OUTPUT : The LED linked to the chosen GPIO pin on the LPC2148 turns on and off one after the other, creating a blinking effect.

CONCLUSION : The LED was connected to the LPC2148 embedded system using a GPIO pin.This experiment shows how GPIO pins can be set up and managed as digital output signals to control external components like an LED.By using the correct voltage levels and adding suitable delays, the LED was able to be switched on and off successfully.